

Original papers

Medial axis-based machine-vision system for orchard robot navigation

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ABSTRACT

In this paper, we propose a novel medial axis-based machine vision technique for autonomous path tracking and navigation of an agricultural robot in orchards. In the proposed method, the raw color image captured by the onboard camera is converted to grayscale using the color index of vegetation extraction. The gray image is run through Gabor filters to extract the texture features before applying Principal Component Analysis technique to minimize the computational load during path coordinate extraction. The resulting image is clustered using K-means clustering algorithm with $k = 2$ as proved best based on Silhouette method. Navigation path is then extracted using medial axis algorithm from the binary image generated after k-means clustering. Fuzzy logic controller with two inputs, heading and offset, is used to smoothly track the medial axis during navigation. Experiment is carried out in an orchard environment to gauge the performance of the system. The results of field experiment clearly demonstrate that the proposed medial axis technique has the potential to accurately extract guidance path for robot navigation. The navigation performance of the system is quite satisfactory with a maximum trajectory tracking error and standard deviation of 14.6 mm and 6.8 mm respectively. The average root mean square error (RMSE) for the lateral deviation was 45.3 mm.

1. Introduction

The global food demand is expected to increase exponentially due to the ever-growing world population, which is projected to about 9.6 to 12.3 billion people by the year 2100 (Gerland et al., 2014). Therefore, as a counter-measure against global food insecurity, the current agricultural production capacity must be doubled by 2050 (Foley et al., 2011). Intensive and extensive farming has long been adopted as a solution. However, this practice suffers a high production cost and requires a large labor force (Van der Ploeg, 2000). Unfortunately, the most dependable energetic rural labor force is declining due to rural-urban migration (Hassall, 2010). With the development of technology, Precision farming (PF) has been applied to optimize production by minimizing costs while maximizing production as well as increasing the work efficiency in intensive and extensive agriculture (Graf Plessen, 2019; Lee et al., 2010). Additionally, agricultural production has been reported to have significantly increased over the years due to intensification, mechanization, and automation of agricultural operations (Nof, 2009; Opiyo et al., 2020; Zhang, 2013).

Current researches are focused on developing and implementing

robotics in agricultural fields as sustainable production systems to ensure constant production against the influencing limiting factors (Burks et al., 2008). Researchers claim that approximately 80% of a task is easy to adapt to robotics or automation, with the remaining 20% considered to be difficult (Stentz et al., 2002). Hence, by adopting the use of robots on the easy parts of a task, the required manual work can be reduced by 80%. However, the application of these robotic systems in agriculture faces significant challenges, especially in navigation due to the unstructured nature of the environment with lots of complex dynamics including wheel slippage, difficulty in segmentation caused by feature similarity between foreground and background, sensor occlusion as a result of weather conditions, vast coverage area among others (Blackmore et al., 2005). This is in contrast to industrial robotic applications where a controlled environment is guaranteed (Hvilshøj et al., 2012).

In orchards, an autonomous robot can be designed to perform multiple tasks, with greater efficiency and precision, such as; yield estimation (Underwood et al., 2016), spraying (Blanco et al., 2019), pruning (Karkee et al., 2014), harvesting (Williams et al., 2019), transplanting (Blackmore et al., 2005), tilling (Yamamoto et al., 2001) among others.

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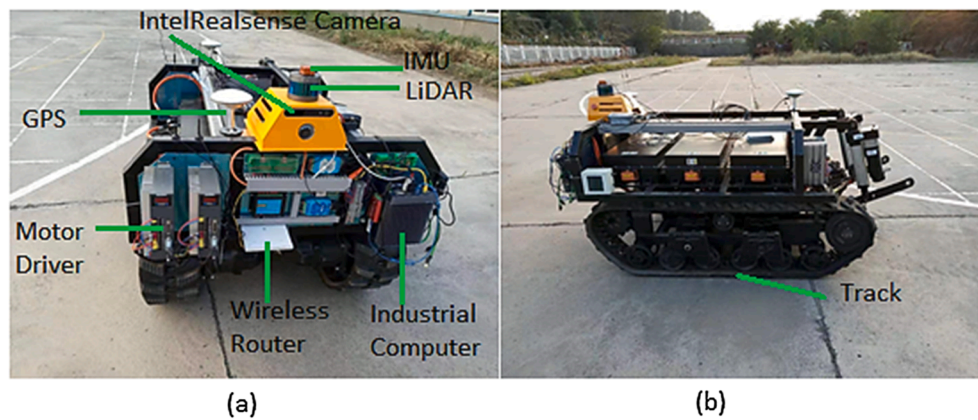


Fig. 1. Tracked orchard robot platform (a) front view (b) side view.

For an effective and efficient task performance, a robot must be able to accurately navigate through the farm while undertaking the task and approaching the target. Therefore, the success of an autonomous agricultural robot greatly depends on the effectiveness, robustness and accuracy of the navigation system.

Varies techniques for robot navigation systems have been proposed by different researchers, such as; global positioning systems (GPS) (Li et al., 2018; Qifeng et al., 2019), dead reckoning (Deng et al., 2018; Dong et al., 2011), radio frequency identification tags (Ampatzidis et al., 2009; Sahba and Nourani, 2016), laser radar (Malavazi et al., 2018), Kalman filter (Blok et al., 2019), and machine vision (Guijarro et al., 2011; Hague et al., 2000; Sergiyenko and Tyrsa, 2020). GPS is one of the most common types of navigation sensors since it can render absolute position measurements to be used by an autonomous robot for agricultural field navigation and task achievement. Bell (2000) developed a four-antenna real-time kinematic GPS (RTK-GPS) based navigation system with a tracking accuracy of between 0.04 and 0.06 m at typical operating speeds. Stoll and Kutzbach (2000) explored the use of only RTK-GPS to steer an agricultural vehicle automatically. To provide more accurate navigation information, the GPS sensor was fused with other sensors in the study by Hellström (2002). Eaton et al. (2008) combined RTK-GPS with inertial measurement unit (IMU). Nørremark et al. (2008) integrated RTK-GPS and tilt sensor to develop autonomous system for intra-row weed control. However, the main challenge to the application of GPS for navigation is the obstruction of line-of-sight to satellites and interference from other Radio Frequency (RF) sources (Hamner et al., 2010). Additionally, based on the structure of an orchard, GPS may not be effectively applied for localization and navigation in some orchards since the agricultural robots frequently move under the tree canopy, which blocks the satellite signals to the GPS receiver (Kise et al., 2001; Li et al., 2009; Niewola, 2020). Therefore, many researchers are exploring autonomous navigation systems for mobile orchards robots without using GPS as the primary sensor for navigation.

Machine-vision systems are capable of detecting navigation paths based on crop rows. Thus a robust vision-based navigation system can be developed to guide autonomous vehicles between crop rows as they undertake field tasks (Benson et al., 2003; Tillett and Hague, 1999; Tillett et al., 2002). Several researchers have explored machine vision techniques to generate navigation paths for autonomous robots. Hague and Tillett (2001) presented a method using a bandpass filter to diminish the grey levels of weeds and shadows in the captured images from the field. A segmentation free crop row detection and localization method of low computational load was proposed by Søgaard and Olsen (2003). Other computer vision-based approaches aimed at generating navigation path between tree rows by detecting the tree canopy and or stems have also been explored. A machine-vision-based orchard navigation path generation algorithm for harvesting robot was developed by He et al. (2010) using horizontal projection to recognize the trunks area

automatically. The algorithm yielded a tree recognition rate of 91.6% and a satisfactory navigation path. A classification-based tree trunk detection method for autonomous navigation was explored by Ali et al. (2008). Additionally, Radcliffe et al. (2018) developed a machine vision navigation system for a small-unmanned ground vehicle in an orchard using the sky and the canopy as the primary focus for navigation path generation. This technique avoided the effect of weeds (color) during segmentation. Ponnambalam et al. (2020) studied an adaptive multi-ROI method aimed at fitting trajectories within drivable regions in Strawberry Fields. An open-loop field test approach with a real agricultural robot demonstrated favorable comparability of their approach to other traditional path generation approaches. Under vision based navigation systems, camera-based systems are the most cost effective technique in comparison to other techniques like LIDAR and Radar based systems. (Islam et al., 2018).

Developing a real-time fully functional navigation system for autonomous guidance is quite a grueling task due to the dynamic nature of agricultural fields for instance, non-uniform terrain, weeds conditions, changing weather conditions, varied illumination among others (Hague et al., 2000; Shalal et al., 2013). In vision-based navigation systems, these factors make it very cumbersome to segment the scene, which is the first and critical step in path generation (Chen et al., 2020). Moreover, in several related studies the developed algorithms first extracted the tree rows (He et al., 2010) before extracting the guidance path. In contrast, the proposed algorithm in our work only uses the binary image and extracts the navigation path hence reducing the computation load on the system. Motivated by these factors, this study is aimed at developing a novel machine vision system for autonomous inter-row navigation for agricultural robot based on medial axis detection. We deviate from the conventional RGB segmentation and employ Gabor filter banks alongside principal component analysis techniques. The main objective of the current study is to generate an accurate navigation path for an autonomous robot in a dynamic orchard setting. Specific objectives include; a) to develop effective, robust, and accurate path segmentation based on Gabor filter, b) To extract the region of interest (ROI) from the segmented path and c) To extract the medial axis for robot navigation.

This paper is organized as follows: Section 2 details the materials and methods employed to achieve the research objective. In Section 3, experimental results and discussion on the proposed technique are presented. Finally, in Section 4, the conclusion and future work is provided.

2. Material and methods

2.1. Vehicle platform

This work employed the use of a tracked orchard robot designed and

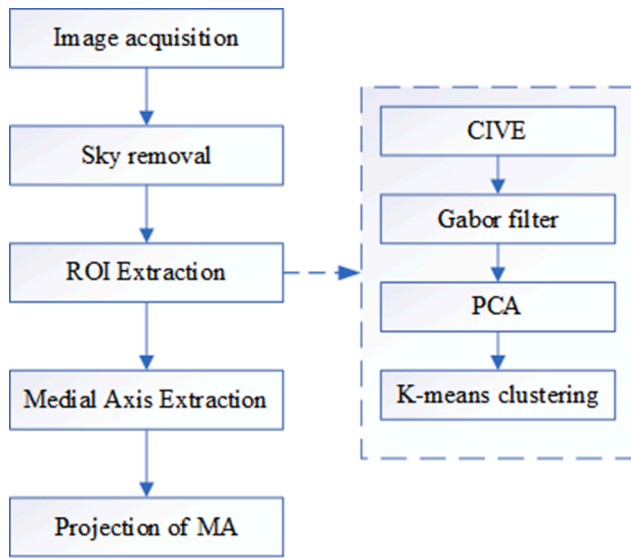


Fig. 2. Flow diagram of image acquisition and processing approach.

developed at the college of engineering of Nanjing Agricultural University, Jiangsu Province, Nanjing China. The platform consists of tracks and uses skid steering technique. Two driving motors (AC Servo Motor 60ST-M01930, 380 V) were separately attached to drive the left and the right tracks via servo drive (ASDD-30A) from Taizhou Yifeng Electronics co. LTD. The dimensions of the platform are; 2.60 m long 1.25 m wide and 1.20 m height. To enable smooth operation in an orchard bearing in mind the rugged terrain, the platform is raised 0.40 m above the ground. The weight of the robot, 700 kg, is well spread over the entire platform, and the wide base ensures stability of the vehicle platform. Fig. 1 shows the vehicle platform used in this work. The central onboard processor is an industrial computer with an Intel Core i5, 2.4 GHz CPU with 8 GB DDR3 RAM. The software for passing the control signals is Robot Operating System (ROS) (Quigley et al., 2009) running on top of Ubuntu Linux version 16.04 operating system installed on the industrial computer. The Intel® RealSense™ Camera D435 RGB channel was used for orchard image acquisition.

2.2. Motion control

This study adopted a skid steering control for the autonomous maneuver of the robot in the orchard. In skid control, steering is achieved through the difference between the velocity of the left and the right motors that drive the tracks/wheels. The error between the left and the right motor speeds results in a change of the vehicle heading, with the yaw angle directly varying with the magnitude of the velocity difference. In case the two velocities are equal, no directional change

happens, and the vehicle continues with no lateral deviation. The steering angle is achieved through a fuzzy controller algorithm developed and refined following numerous laboratory and field tests. Unlike proportional-integral (PI) and proportional integral derivative (PID) controllers, fuzzy controllers can be designed to guarantee stable operation despite the change in parameters or motors (Kumar et al., 2004). Fuzzy controllers consist of three main parts; fuzzification, fuzzy reasoning, and defuzzification. These parts represent the input stage, the processing stage, and an output stage. At the input stage, inputs or errors are mapped to the relevant membership functions and truth values. The processing stage invokes each appropriate rule and generates the rules. The output stage transforms the combined result back into a specific control output value. For good performance of the controller the number of parameters should be carefully selected (Vans et al., 2014). In this study, after determination of the offset and the heading of the robot in relation to the medial axis, the velocities of the left and the right tracks were adjusted to steer the robot towards the medial axis (coordinates). The fuzzy controller with two inputs, offset and heading of the robot, was implemented according to (Xue et al., 2012).

2.3. Image acquisition system

This experiment was performed in a commercial apple orchard in Jiangsu province, Nanjing China. IntelRealSense D435 camera was used to acquire RGB images at a frame rate of 1 fps and a resolution of 640×480 . At normal operation, IntelRealSense D435 is capable of acquiring images at up to 30 fps at a resolution of 1920×1080 (Intel-corporation, 2018). However these were adjusted to ease computational load during image processing hence enable the realization of real-time robot control.

The camera was placed at the top center of the robot about 1.2 m from the ground and with its y-axis perpendicular to the world's horizontal plane. The camera location and orientation followed a design that considered the field of view of the sensor and various dynamics of the orchard robot, including the presence of tall grass that may interfere with the camera image. The camera was interfaced with the computer through a USB-3.0 port for improved image acquisition speed. The algorithm flow diagram of the proposed MA-based navigation system is shown in Fig. 2.

2.4. Image pre-processing

2.4.1. Image transformation and smoothing

Raw images acquired from the field are always affected by several factors, including varying background, illumination as well as occlusion hence it is impractical to process the image in its raw form (Bhattacharyya, 2011; Kurka and Díaz Salazar, 2019). In image pre-processing, the general target is to employ various techniques to transform the raw image into a well-structured representation of the environment. Pre-processing may involve one or several steps, depending on the nature of the raw image. In this work we employed two techniques; CIVE (color

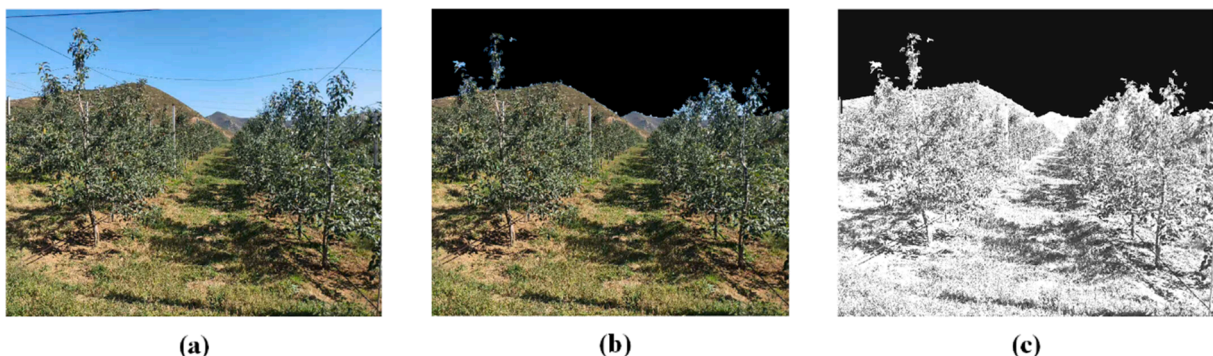


Fig. 3. The image preprocessing steps. (a) Raw RGB image, (b) sky removed image and, (c) CIVE grayscale image.

index of vegetation extraction) (Hamuda et al., 2016) and median filter (Gupta, 2011).

Firstly, a sky removal operation was performed based on pixel threshold technique as given in Eq. (1) to facilitate in the proceeding image processing operations by eliminating unnecessary image objects, thus reducing the computational cost.

$$I_{x,y} = \begin{cases} 0 & \tau_{min} < A_{x,y} < \tau_{max} \\ A_{x,y} & \text{otherwise} \end{cases} \quad (1)$$

Where $I_{x,y}$ is the resultant image after sky removal, $A_{x,y}$ is the original image, τ_{min} and τ_{max} are the minimum and maximum pixel values in $L*a*b*$ color space.

Secondly, image, $I_{x,y}$, was converted to CIVE according to Eq. (2). For

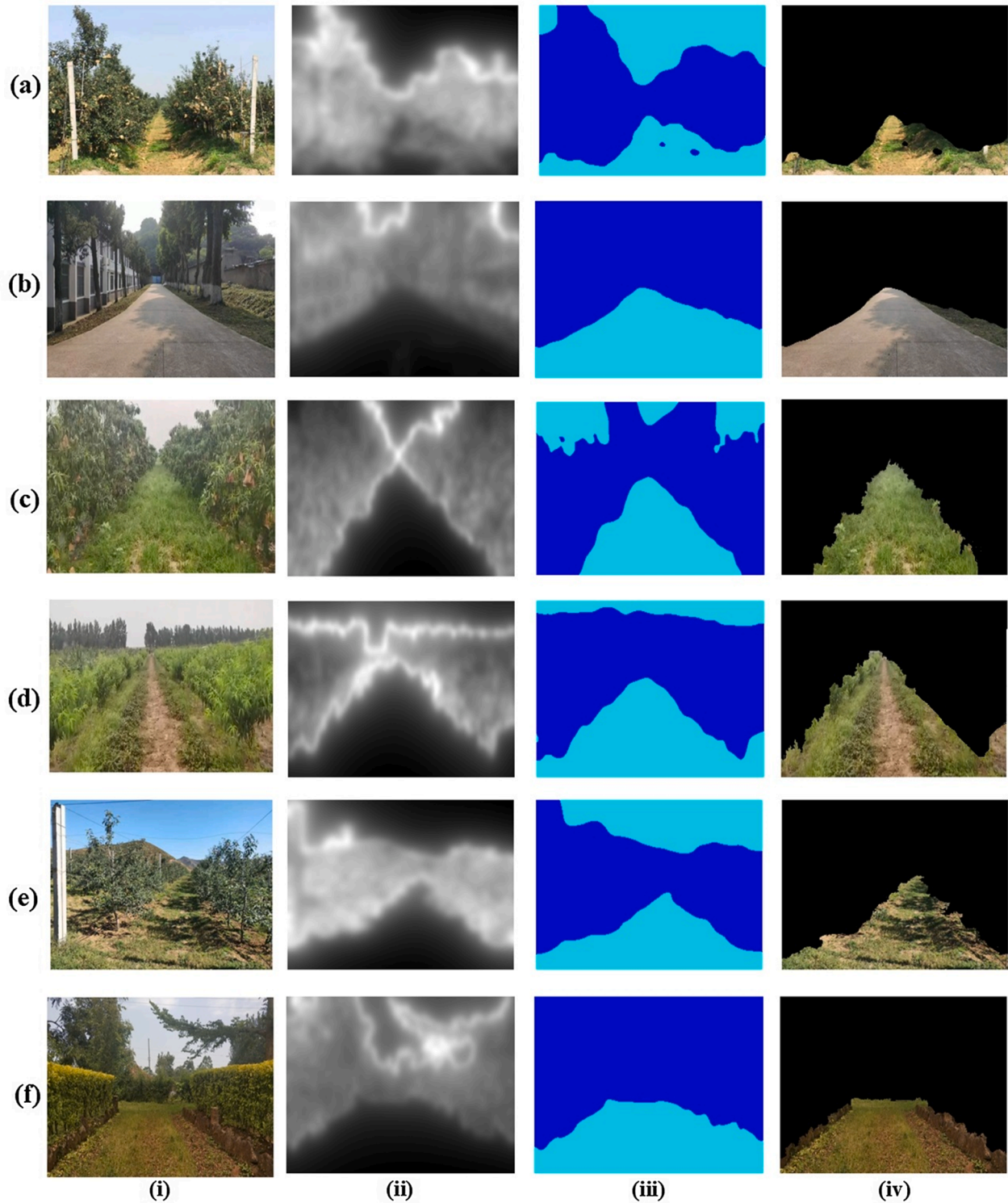


Fig. 4. (a)-(c) Display of extracted inter-row from different orchard environments (i) Original Image, (ii) Gabor filter, (iii) k-means cluster and (iv) extracted inter-row path.

every RGB image pixel, the intensity values of the R, G and B channel are converted to grayscale value forming a grayscale image with each pixel represented by CIVE intensity equivalence. Fig. 3 presents the image preprocessing steps from raw RGB image, sky removal and CIVE image.

$$C_{x,y} = 0.441R + 0.811G + 0.385B + 18.7874 \quad (2)$$

Finally, image smoothening (Eq. (3)) was performed by a median filter of 3×3 kernel neighborhood to remove noise in $C_{x,y}$ for subsequent processing. The choice of kernel size is always a tradeoff between the quality of the output and the processing time. In this work 3×3 kernel gave satisfactory quality of output without straining the processing resources. Generally, median filters are non-linear, edge-preserving smoothing methods aimed at eliminating impulse noise in a given image. The working principle of the median filter is based on the replacement of the gray level of every pixel by the median of the gray levels in a neighborhood of the pixels, instead of employing the average operation (Broesch, 2009; Tan, 2013).

$$H_{x,y} = \text{median}\{C_{x,y}[i,j], (i,j) \in k\} \quad (3)$$

Where k represents the neighborhood centered around location $[x,y]$ in $C_{x,y}$.

2.4.2. Image segmentation

Humans can focus on a particular object/point/area of an image regardless of the nature of the surrounding region. However, for computers, this task may be impossible if efforts are not made to extract the ROI from the surrounding region. To enable easy identification of the ROI, image segmentation was implemented. Image segmentation is the process that involves subdividing an image into foreground (required region) and background (non-important region) (Zaitoun and Aqel, 2015) or the process of grouping related pixels together to form connected objects that have relatively homogenous properties (Wang et al., 2019).

2.4.2.1. Gabor filters. Several methods can be adopted for segmentation depending on the problem being solved and the dynamics of the image content. In this work, we utilize the Gabor filter method to extract texture features in an orchard setting. Gabor filter/wavelet is widely used to extract texture features from the images for image retrieval and has been found to be very efficient (Zhang et al., 2000). However, out of the many literature we reviewed (Makem et al., 2020; Mede et al., 2018; Serrano et al., 2010; Shen and Bai, 2006; Zhang and Tan, 2002), none attempted to implement this method in an orchard environment, especially for robot navigation. The closest work that adopted the Gabor filter in a semi-structured environment was by (Alharbi et al., 2018) to develop a classifier for automatic counting of wheat spikes from wheat growth images.

Gabor filters rely on the theory fronted by Campbell and Robson (1968), which holds that the visual system decomposes the retinal image into several filtered images, each containing intensity variations over a narrow range of frequency (size) and orientation. In this work, a bank of 2D Gabor filters was explored. The 2D Gabor function comprises a sinusoidal plane wave with frequency and orientation, which is modulated by a 2D Gaussian envelope. For an image frame $H_{x,y}$ having size $x \times y$ captured by the robot camera, the responsive impulse of a Gabor filter is computed using Eq (4).

$$H'_{x,y} = \exp\left\{-\frac{1}{2}\left(\frac{x^2}{\beta_x^2} + \frac{y^2}{\beta_y^2}\right)\right\} \cos(2\pi\mu_0 x + \phi) \quad (4)$$

where μ_0 and ϕ represent the frequency and phase of the sinusoidal plane wave along the x-axis or 0° orientation, and β_x and β_y represent the space constants of the Gaussian envelope along the x- and y-axis, respectively. Depending on the image size, after the application of the Gabor filter, several features are produced that may cause too much load

during analysis.

2.4.2.2. Principal Component analysis (PCA). For each pixel 34 features were extracted by Gabor filter. Hence, to minimize the computational demand during processing, Principal Component Analysis (PCA) technique was employed. The central idea of PCA is to reduce the dimensionality of a data set having a large number of interrelated variables while retaining as much as possible of the variation present in the data set. PCA generates a new set of orthogonal features known as principal components by transforming several interrelated variables (Jolliffe and Cadima, 2016) hence eliminating redundancies. In this work, Gabor filter data is run through the PCA algorithm, which identifies principal components used as texture descriptors representing the orchard path, thus allowing more natural segmentation with reduced data dimensionality.

2.4.2.3. Data clustering using k-means. The final step in the segmentation stage was to classify the data using the k-means algorithm. K-means is reported as among the simplest unsupervised learning algorithms that solve the well-known clustering problem (Kodinariya and Makwana, 2013; Tian et al., 2019). The procedure adopts a simple way to classify a data set through a certain number of clusters fixed a priori. One of the controversial issues in clustering is the determination of the number of clusters appropriate for the data set. There are several techniques employed in determining the number of clusters, including; the rule of thumb (Madhulatha, 2012), Elbow method (Bholowalia and Kumar, 2014; Umargono et al., 2020), Information Criterion Approach (Bozdogan, 1994), Information-Theoretic Approach (Sugar and James, 2003), Using Silhouette (Kodinariya and Makwana, 2013; Shahapure and Nicholas, 2020) and Cross-validation methods (Smyth, 1996). The silhouette method was selected as it is lighter and gives more reliable results independently (Ossama et al., 2011). It works by measuring how coherent a point is to its cluster in comparison to other clusters based on Silhouette value ($s(i)$) for any point i in $H_{x,y}$ described by Eq (5).

$$s(i) = \frac{b(i) - w(i)}{\max\{b(i), w(i)\}} \quad (5)$$

Where $b(i) = \min_k\{B(i,k)\}$, $w(i)$ represents the average distance from the i^{th} point to the other points in its cluster and, $B(i,k)$ represent the average distance between the i^{th} point to another cluster k . The silhouette values range from -1 to $+1$. A silhouette value of about zero means that the entity could be assigned to another cluster as well. If the silhouette value is close to -1 , the entity is misclassified. Silhouette values close to $+1$, means that the set is well clustered. In this study k was set to 2 based on silhouette evaluation. The sparsity of $I_{x,y}$ (sky removed image) was cloned based on the extracted clusters (k-means clusters) using Eq (6). Fig. 4 shows a series of images from different orchard environments clearly showing how application of Gabor filter and k-means clustering aided in detecting the inter-row path prior to extraction of the MA coordinates.

$$G_{x,y} = \begin{cases} 0 & \text{if } c = 1 \\ I_{x,y} & \text{otherwise} \end{cases} \quad (6)$$

Where $G_{x,y}$ is the resultant $I_{x,y}$ clone, c is the cluster. For this study, cluster $c = 1$ represented the trees, while cluster $c = 2$ represented the path and the sky. Finally, $G_{x,y}$ was binarized by Otsu's method (Otsu, 1979).

K-means clustering was very handy in determining which cluster represented the path as this was our region of interest (ROI). An algorithm was developed that could help the system detect which cluster represented the path. A summary of the steps is outlined hereafter. Firstly the center of the image C_{x_m,y_m} was extracted. Several trials demonstrated that in any given frame, the path was closer to the bottom of the image. In that regard a threshold ω given by Eq. (7) was applied to

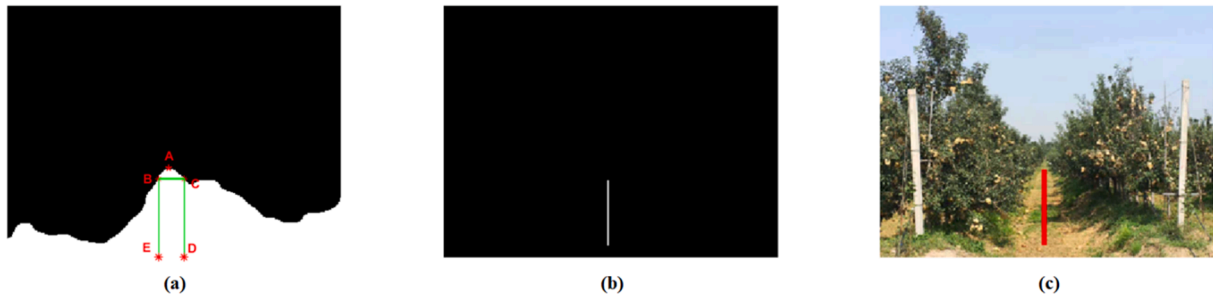


Fig. 5. The medial axis extraction process, (a) path image, (b) Medial axis location, (c) projection of medial axis on the original image.

determine the y coordinate(y_r) of the point of interest along the x_m column. Where y is the height of the image.

$$y_r = 0.9 * y \quad (7)$$

Therefore, the cluster of point of interest R_{x_m, y_r} represented the path's cluster. All these were done considering that the origin of the image is on the top left corner of the image.

The use of Gabor filters made it very easy for the k-means clustering algorithm to properly cluster the inter-row path despite the shadows that appear in some of the images for instance Fig. 6a(i), b(i) and e(i), a perfect demonstration of the effectiveness of integrating CIVE, Gabor filter and k-means for path clustering.

2.5. Image processing

This algorithm aimed at extracting the robot navigation path based

on a shape representation technique of the extracted ROI. Shape description methodologies look for effective and perceptually important shape features based on either contour or interior information (Zhang and Lu, 2004). Medial axis shape representation was applied in this study to extract the navigation path coordinates.

2.5.1. Medial axis extraction

The medial axis transform (MAT) is a powerful shape abstraction which renders a compact representation that preserves topological properties of the input shape. These properties are invariant to translation, rotation, scaling, articulation while their locality is robust to occlusion (Tsogkas and Dickinson, 2017).

Several researchers have reported the feasibility of MAT to effectively reduce computational complexity of algorithms for a span of tasks including mesh editing (Yang et al., 2018; Yoshizawa et al., 2003), shape recognition (Shen et al., 2016) and matching (Yasseen et al., 2016) as

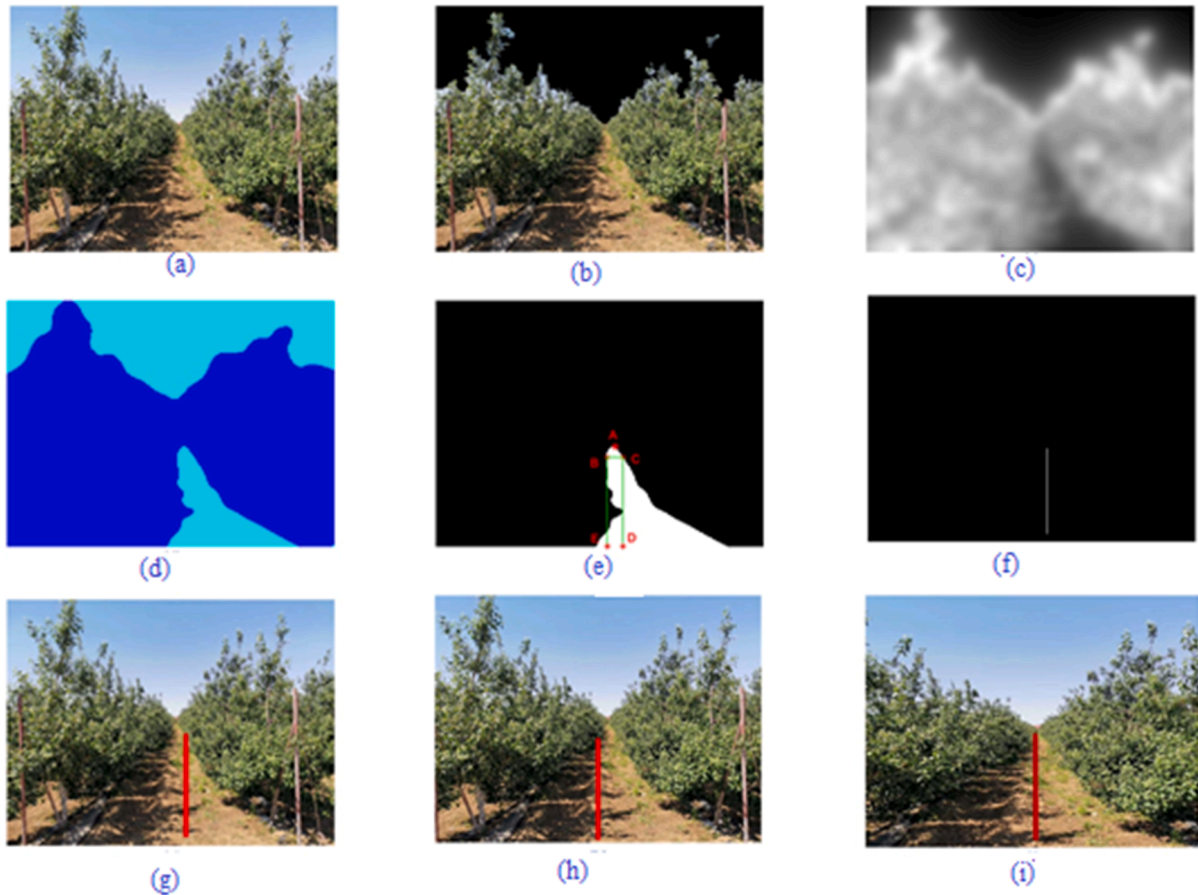


Fig. 6. Image processing and medial axis extraction (a) Original image (b) removal of the sky (c) Gabor filter application (d) k-mean clustering (e) binary image (f) extracted MA (g) projection of medial axis on the original image (h/i) successive paths generated as the robot moves forward.

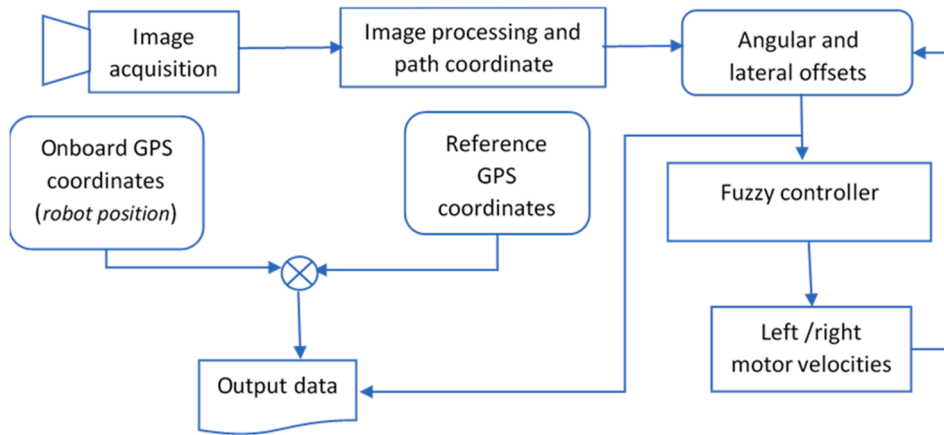


Fig. 7. Control loop diagram for robot motion.

well as shape manipulation (Du and Qin, 2004). However, no studies have been reported on the application of MAT for robot navigation path extraction.

The Medial axis (MA) is a set of centroids of maximally inscribed disks contained in a shape and are tangential to the shapes boundary in at least two points (Blum, 1967). The MA together with the values of the radii of these maximally inscribed disks is called the MAT. In this study, only the MA (x and y centroid coordinates) were applied as the navigation path. The procedure for MA extraction is given below.

Step 1: Detect the path end point G_{ij} by extracting the image object point with the least y-coordinate value, as shown in Fig. 5(a) with the letter A, where i and j are the x and y coordinates of the point A.

Step 2: Detect the path edges, $G_{m,n}$ and $G_{p,n}$ represented by points B and C, respectively. If the image size for $G_{x,y}$ is x by y , and considering that the origin of the image is on the top left corner, the path length (p_l) was computed by $p_l = y - j$. The y-coordinate value n for $G_{m,n}$ and $G_{p,n}$ was determined by a set threshold (Γ) from the point G_{ij} as indicated in equation (8).

$$\Gamma = 0.15 * p_l \quad (8)$$

Γ was set based on several trials to obtain an optimum edge position. The x-coordinate values m and p were determined as the first and the last ROI pixel positions respectively, along $y = n$.

Step 3: Inscribe a rectangle (B, C, D, E), whose length and width are determined by p_l and edges (B, C), respectively, as shown in Fig. 5(a) (green lines).

Step 4: Compute MA (Fig. 5(b)) based on the image thinning technique (Lam et al., 1992). According to Lam et al. (1992), this technique of medial axis extraction offers a strong representation while retaining only the essential skeleton branches.

Step 5: Finally, project the extracted medial axis to the path end by concatenating the MA with point A coordinates, as shown in Fig. 5(c).

To check the robustness of the developed algorithm, we tested the

system in various sections of the orchard. Fig. 6 shows the output of the medial axis for a different part of the orchard. It can be observed in Fig. 6 (e) that the binary image tends to portray the existence of a false obstacle due to the robot's orientation and shadow. Interestingly, the medial axis extraction algorithm was able to extract the robot path as shown in Fig. 6(g). We must mention that the robot path was generated continuously from successive frames at a rate of 1 fps. In this case, any previous path error can be corrected in subsequent frames as demonstrated in Fig. 6(g), 6(h) and 6(i). This therefore demonstrate the robustness of the algorithm.

2.6. Experimental results and discussions

To verify the performance of the proposed vision-based path tracking system, field test was carried out using the tracked robot described in Section 2.1 above. The proposed system stability, efficiency and robustness was determined by lateral (ψ) and angular (θ) deviations evaluation. Therefore, two sets of experiments were carried out to determine the performance of the proposed system with reference to ψ and θ of the robot. Note that in all the field tests the linear forward velocity of the robot was set to 0.2 ms^{-1} which was the optimum velocity that resulted in most stable motion. Moreover, the experiments were conducted on a clear day between 1000 h and 1500 h China time (GMT + 8).

2.7. Lateral deviation evaluation

ψ is characterized by either positive (left of center line), negative (right of center line) or zero (at the center line) deviation with reference to the center line position. In this test, five rows ($R = \{R_1, R_2, R_3, R_4, R_5\}$) were randomly selected and the robot was placed at the beginning of each row at varying deviations of 10.5 mm, 13.7 mm and 17.2 mm for positive $+\psi$ and 13.2 mm, 8.7 mm and 7.8 mm for

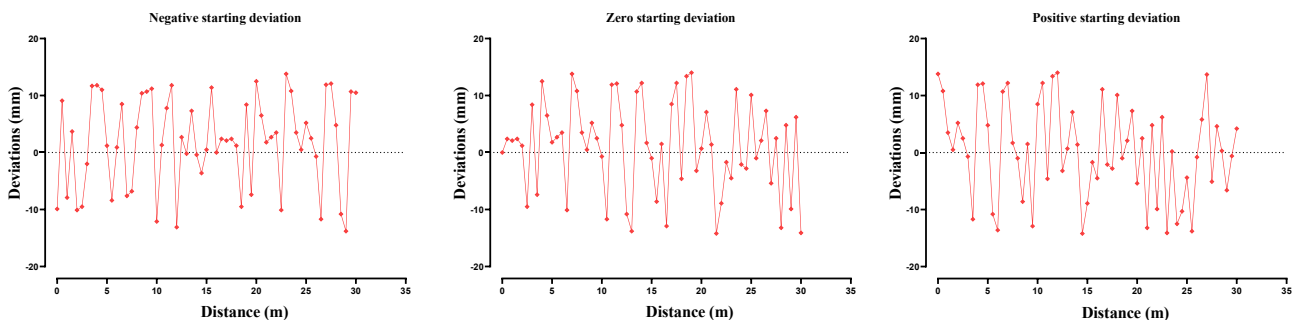


Fig. 8. Trajectory tracking errors for variable starting deviations.

Table 1
Proposed guidance system performance for selected rows.

Row(R)	Lateral deviation(ψ)			
	Maximum (mm)	Average (mm)	RMSE(mm)	std
R ₁	12.5	3.5	48.2	6.6
R ₂	-13.8	-4.2	67.1	6.8
R ₃	-13.2	-2.7	35.7	5.9
R ₄	14.6	-2.9	42.2	6.5
R ₅	11.7	3.3	33.2	6.6

Where std is standard deviation.

Table 2
Mean and standard deviation for varied starting deviations.

$-\psi$ (mean $\pm$ std)	ψ_0 (mean $\pm$ std)	$+\psi$ (mean $\pm$ std)
6.892 $\pm$ 4.774	6.692 $\pm$ 4.640	6.931 $\pm$ 4.377

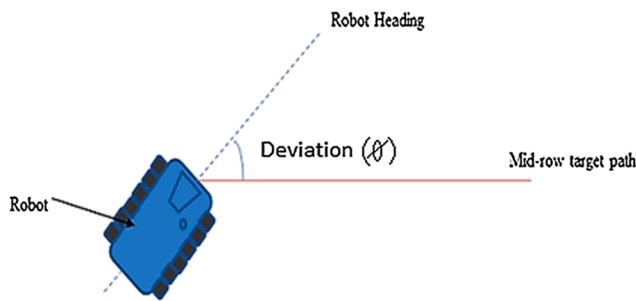


Fig. 9. Robot angular orientation.

negative $-\psi$ and at zero ψ_0 (varying deviations for three replications). The RTK - GPS data was applied as the ground truth path trajectory to validate the performance of the proposed system. Therefore, the trajectory error (ψ) was determined from the difference between the centerline and the robot path as given by the RTK - GPS data. All the experiments were conducted under clear sky and the orchard canopy did not overlap across the rows at any given point. Moreover, the fusion of IMU and GPS using Kalman Filter (KF) was implemented to improve the accuracy of the obtained data (Sun et al., 2019). A block diagram of the control loop for the robot navigation is shown in Fig. 7 below.

In this test, the robot was placed at the beginning of each R at the ψ values then the robot was autonomously set to move through the Rs for a distance of 30 m. The Fig. 8 shows the average ψ for $+\psi$, zero ψ , and $-\psi$ starting deviations along the 30 m autonomous movement along the Rs. It was established that in all the three cases ($+\psi$, zero ψ , and $-\psi$), the absolute deviations are within 15 mm, demonstrating the robustness of the system to varied starting lateral errors.

Table 1 presents the performance of the introduced system in terms of trajectory errors from the center line. The proposed system returned a maximum ψ of -13.8 mm and 14.6 mm in R₂ and R₄, respectively. This indicates that the robot could steadily traverse between the rows without affecting the orchard trees.

Additionally, the std was relatively constant ranging between 5.9 and 6.8 for all the Rs, this further indicates the reliability of the introduced vision system and effectiveness of the fuzzy controller leading to stability of the robot during navigation. The high RMSE in R₂ was due to slightly rough ground in the row as well as wider gaps between tree canopies (in comparison to other Rs), which influenced the medial axis extraction process. The average RMSE for the lateral deviation was 48.3 mm.

A statistical test based on the dependent *t*-test between ψ_0 against $+\psi$, and ψ_0 against $-\psi$ was performed. It was established that there was no statistical ($p < 0.0005$) significance differences between $-\psi$ (mean $\pm$ std) and ψ_0 (mean $\pm$ std) nor between $+\psi$ (mean $\pm$ std) and ψ_0 (mean $\pm$ std). This was attributed to the ability of the robot camera to capture the orchard row almost similarly despite the starting lateral deviations. Table 2 presents *t*-test outcome for the varied starting deviations.

2.8. Angular deviation evaluation

θ evaluation was aimed at determining the robustness of the system against variation in camera angular deviation (θ). The robot was started at the beginning of each R at varied orientations of -40° to $+40^\circ$ with respect to the guidance path as shown in Fig. 9. Note that deviation of more than $\pm 40^\circ$, failed to generate guidance path coordinates due to relatively narrow lateral field of view of the camera ($\sim 81^\circ$) thus are not included in this work. It is equally worth noting that in this work, we represented orientation as the angular deviation of the robot from the MA guidance path.

The Zero-degree deviation coincided with the target path and the orientation steps were set to 10° . At different starting deviation, the robot's angular adjustment towards the target path was recorded and presented graphically in Fig. 11. In all the cases the robot was placed at the midline of the orchard row, hence, the lateral deviation was zero.

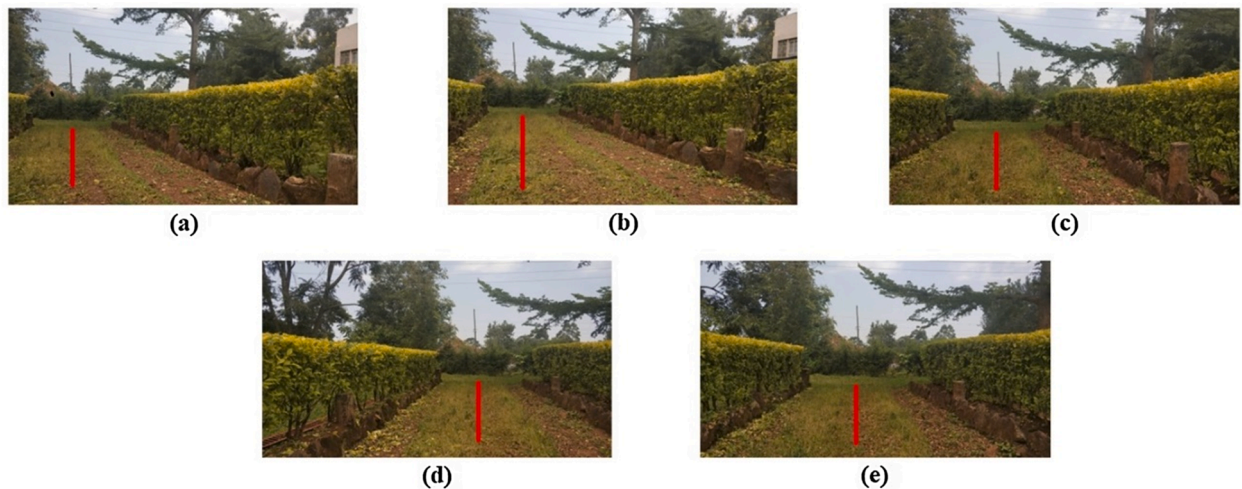


Fig. 10. Series of images for robot navigation at a deviation of 400: (a) initial orientation (b) robot tending to turn left slightly (c) robot makes increased turn to the right (d) robot makes slight turn to the left (e) The robot finally aligns to the inter-row path.

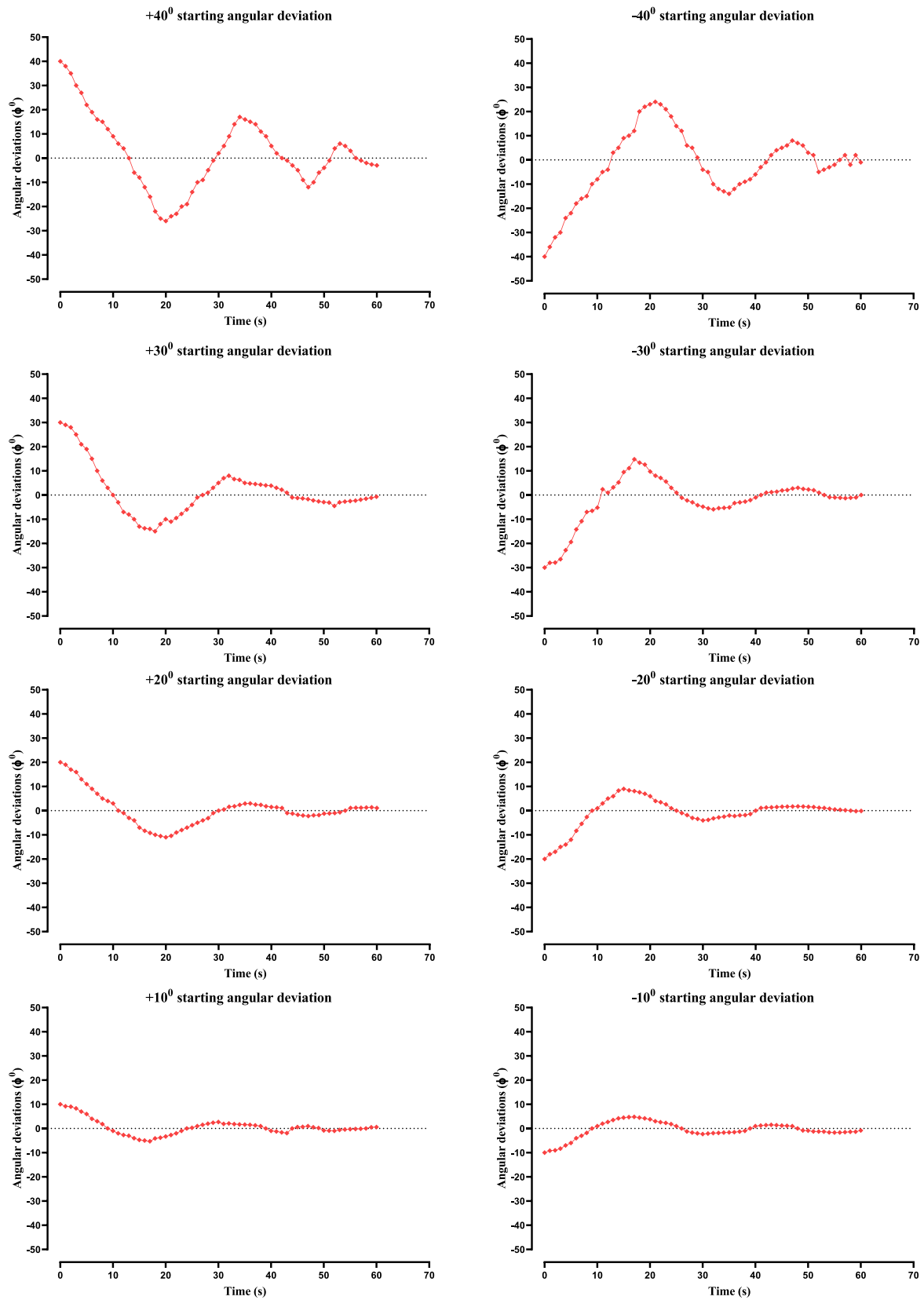


Fig. 11. Tracking guidance line from different starting orientations.

Table 3

Statistical significance difference between individual deviation groups.

θ_1 (mean $\pm$ std)	θ_2 (mean $\pm$ std)	θ_3 (mean $\pm$ std)	θ_4 (mean $\pm$ std)	θ_{-1} (mean $\pm$ std)	θ_{-2} (mean $\pm$ std)	θ_{-3} (mean $\pm$ std)	θ_{-4} (mean $\pm$ std)
2.292 $\pm$ 0.279 ^d	4.713 $\pm$ 0.344 ^c	7.165 $\pm$ 0.361 ^b	11.633 $\pm$ 0.409 ^a	2.529 $\pm$ 0.239 ^{de}	4.083 $\pm$ 0.347 ^{ce}	6.361 $\pm$ 0.382 ^{bc}	10.803 $\pm$ 0.364 ^a

a,b,c mean $\pm$ std within a row, with no superscript in common differ significantly ($p < 0.05$).

An efficient orientation adjustment is very critical as it indicates how well the onboard camera would quickly balance the view of the row, thus, affecting the subsequent path generation process. In this experiment the stability threshold was set at $\pm 5^\circ$ based on the nature of the terrain of the orchard. It was experimentally confirmed that the robot was able to maintain its orientation within the set threshold.

A larger starting θ ($\pm 40^\circ$) showed much difficulty in path tracking with initial absolute angular overshoot of about 25° . At this starting angle the robot gained stability after about 56 s. We selected a series of images (Fig. 10) obtained in a row as the robot tried to detect the path at an angular deviation of 40° . The images clearly indicate the difficulty of the robot to capture detect the correct path initially. However, as it moves forward it tends to correct the path in successive frames until it finally gets the correct path as in Fig. 10(e).

For a starting $\theta \pm 30^\circ$ the angular adjustment was relatively smoother with an initial absolute overshoot of about $\pm 14^\circ$ and stability after about 40 s. Even though the robot maintained its stability after 40 s, there was a slight deviation at 53 s which may have been due to a slip. The sharp peaks observed for starting θ of $\pm 30^\circ$ and $\pm 40^\circ$ could have been as a result of camera not being able to capture one row sufficiently thus generating noisy medial axis coordinates. Quicker stability achievement is observed for starting θ of $\pm 20^\circ$ and $\pm 10^\circ$. Other than the first overshoot in $\pm 20^\circ$ starting θ , the rest of the deviation were within the stability threshold. The lowest starting deviation of $\pm 10^\circ$ indicated less struggle by the robot to track the navigation path. This can be attributed to a better view of the rows by the onboard camera thus a quicker and accurate medial axis generation.

A one-way analysis of variance (ANOVA) was performed to determine the differences in the path navigation based on the different starting θ i.e., θ_{-4} , θ_{-3} , θ_{-2} , θ_{-1} , θ_1 , θ_2 , θ_3 , and θ_4 for -40° , -30° , -20° , -10° , $+40^\circ$, $+30^\circ$, $+20^\circ$, and $+10^\circ$, respectively. There was a statistically significant difference ($p < 0.05$) between starting θ groupings. A Tukey post hoc test was performed to determine the statistical significance difference between individual starting θ groups as presented in Table 3. θ_1 (mean $\pm$ std) was statistically significantly the lowest. However, there was no statistically significant difference ($p < 0.05$) between θ_1 (mean $\pm$ std) and θ_{-1} (mean $\pm$ std). Thus, path tracking at a starting θ of $\pm 10^\circ$ was relatively the same irrespective of the direction. There was no statistically significant difference ($p < 0.05$) between θ_2 (mean $\pm$ std), θ_{-2} (mean $\pm$ std), and θ_{-3} (mean $\pm$ std). Additionally, there was no statistically significant difference ($p < 0.05$) between θ_3 (mean $\pm$ std), and θ_{-3} (mean $\pm$ std). Furthermore, there was also no statistically significant difference ($p < 0.05$) between θ_{-1} (mean $\pm$ std), and θ_{-2} (mean $\pm$ std) which may have been as a result of much clearer path vision during the start at θ_{-2} . θ_4 (mean $\pm$ std) was statistically significantly the highest. However, there was no statistically significant difference ($p < 0.05$) between θ_4 (mean $\pm$ std), and θ_{-4} (mean $\pm$ std).

From Table 3 the introduced vision-based path extraction technique, the initial orientation of the robot/camera affect the tracking efficiency of the generated path. This can be of significance particularly in changing of rows (headland turning), in which case the trajectory of the robot should be in such a way that it immediately faces the next row at a small angular deviation thus ensuring smoother motion. Keyhole turning mechanism (Takai et al., 2013) could be an appropriate method to integrate with this vision based navigation.

Though we acknowledge that the experimental field may not be comparable based on their dynamics, we are still convinced that, in the

Table 4

Comparison of maximum deviation for this study with related work.

Study	Area of application	Maximum error (mm)
This work (MA)	Orchard	14.6
(Xue et al., 2012)	Medium height corn field	15.8
(Radcliffe et al., 2018)	Orchard	35.0
(Xue and Xu, 2010)	vegetable field	25.0
(Jones et al., 2019)	Kiwi orchard	75.0
(Subramanian and Burks, 2005)	Citrus Orchard	60.0

scientific field, comparison of the performance of the current work with selected related work in the literature is necessary. In this work, the comparison was based on the maximum lateral deviation, which is critical parameter to demonstrate the ability of the robot to accurately navigate narrow rows and avoid running over crops/trees in the agricultural field. Table 4 shows the comparison of the current work with other related studies.

It can be observed in Tables 1 and 4 that this work (medial axis technique) yielded better results in terms of path tracking with the lowest maximum error of 14.6 mm. We attributed this to the robustness of the medial axis technique developed, well-tuned fuzzy controller and stability as well as high traction efficiency of the tracked robot (Takai et al., 2010) employed in this study.

2.9. Conclusion and future work

A machine vision system based on medial axis extraction technique for orchard robot navigation is successfully developed in this study. The work is presented in two major categories; Image processing to extract the medial axis and trajectory tracking based on the medial axis coordinates. The image processing part consists color index of vegetation extraction (CIVE) for gray scaling, Gabor filters, Principal Component Analysis (PCA), K-mean clustering and medial axis algorithm. Based on the results presented, this technique performed well in segmenting the image and extracting the navigation path coordinates. A two-input fuzzy logic controller is used to track the medial axis smoothly. A skid steering tracked orchard robot is used in the field experiment to test the performance of the developed system. The field test in an orchard environment demonstrates the viability of the system with a maximum path tracking error and standard deviation of 14.6 mm and 6.8 mm respectively. A comparison of this work with previous research based on maximum trajectory error shows better performance in favor of this work. These results demonstrated that the medial axis technique has the potential to generate accurate guidance path for orchard robot navigation. The future work will focus on integrating medial axis technique with laser data to improve path tracking as well as obstacle detection and avoidance. Additionally, augmenting the medial axis on the operator's view during robot teleoperation is another viable area that our subsequent work will focus on.

CRediT authorship contribution statement

Samwel Opiyo: Writing - review & editing. **Cedric Okinda:** Writing - review & editing. **Jun Zhou:** Writing - review & editing. **Emmy Mwangi:** Writing - review & editing. **Nelson Makange:** Writing - review

& editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- Alharbi, N., Zhou, J., Wang, W., 2018. Automatic counting of wheat spikes from wheat growth images. In: 7th International Conference on Pattern Recognition Applications and Methods, pp. 346–355. <https://doi.org/10.5220/0006580403460355>.
- Ali, W., Georgsson, F., Hellstrom, T., 2008. Visual tree detection for autonomous navigation in forest environment. 2008 IEEE Intelligent Vehicles Symposium, pp. 560–565. <https://doi.org/doi:10.1109/IVS.2008.4621315>.
- Ampatzidis, Y.G., Vougioukas, S.G., Bochtis, D.D., Tsatsarelis, C.A., 2009. A yield mapping system for hand-harvested fruits based on RFID and GPS location technologies: field testing. *Precis. Agric.* 10 (1), 63–72 <https://doi.org/https://doi.org/10.1007/s11119-008->.
- Bell, T., 2000. Automatic tractor guidance using carrier-phase differential GPS. *Comput. Electron. Agric.* 25 (1–2), 53–66. [https://doi.org/10.1016/S0168-1699\(99\)00055-1](https://doi.org/10.1016/S0168-1699(99)00055-1).
- Benson, E.R., Reid, J.F., Zhang, Q., 2003. Machine vision-based guidance system for agricultural grain harvesters using cut-edge detection. *Biosyst. Eng.* 86 (4), 389–398. <https://doi.org/10.1016/j.biosystemseng.2003.07.002>.
- Bhattacharyya, S., 2011. A brief survey of color image preprocessing and segmentation techniques. *J. Pattern Recognit. Res.* 1 (1), 120–129. <https://doi.org/10.13176/11.191>.
- Bholowalia, P., Kumar, A., 2014. EBK-means: A clustering technique based on elbow method and k-means in WSN. *Int. J. Comput. Appl.* 105 (9).
- Blackmore, S., Stout, B., Wang, M., Runov, B., 2005. Robotic agriculture—the future of agricultural mechanisation. In: *Proceedings of the 5th European Conference on Precision Agriculture*, pp. 621–628.
- Blanco, M.N., Fenske, R.A., Kasner, E.J., Yost, M.G., Seto, E., Austin, E., 2019. Real-Time Monitoring of Spray Drift from Three Different Orchard Sprayers. *Chemosphere* 222, 46–55. <https://doi.org/10.1016/j.chemosphere.2019.01.092>.
- Blok, P.M., van Boheemen, K., van Evert, F.K., Jsselmuiden, J., Kim, G.-H., 2019. Robot navigation in orchards with localization based on Particle filter and Kalman filter. *Comput. Electron. Agric.* 157, 261–269. <https://doi.org/10.1016/j.compag.2018.12.046>.
- Blum, H., 1967. A transformation for extracting new descriptors of shape. *Models for the Perception of Speech and Visual Form* 19(5), 362–380. Retrieved from <http://pageperso.lif.univ-mrs.fr/~edouard.thiel/rech/1967-blum.pdf>.
- Bozdogan, H., 1994. Mixture-model cluster analysis using model selection criteria and a new informational measure of complexity. In: *Proceedings of the First US/Japan Conference on the Frontiers of Statistical Modeling: An Informational Approach*, pp. 69–113. https://doi.org/10.1007/978-94-011-0800-3_3.
- Broesch, J. D., 2009. Applications of DSP. In: J. D. B. T.-D. S. P. Broesch (Ed.), *Instant Access*, pp. 125–134. <https://doi.org/10.1016/B978-0-7506-8976-2.00007-9>.
- Burks, T.F., Schmoldt, D.L., Steiner, J.J., 2008. US specialty crops at a crossroad: Hi-tech or else? *Resource Magazine* 15 (6), 5–6. <https://doi.org/10.13031/2013.29280>.
- Campbell, F.W., Robson, J.G., 1968. Application of Fourier analysis to the visibility of gratings. *J. Physiol.* 197 (3), 551. <https://doi.org/10.1113/jphysiol.1968.sp008574>.
- Chen, J., Qiang, H., Wu, J., Xu, G., Wang, Z., Liu, X., 2020. Extracting the navigation path of atomato-cucumbergreenhouse robot based on a median point Hough transform. *Comput. Electron. Agric.* 174, 105472. <https://doi.org/10.1016/j.compag.2020.105472>.
- Deng, Y., Shan, Y., Gong, Z., Chen, L., 2018. Large-scale navigation method for autonomous mobile robot based on fusion of GPS and lidar SLAM. *Chinese Automation Congress (CAC) 2018*, 3145–3148. <https://doi.org/10.1109/CAC.2018.8623646>.
- Dong, F., Heinemann, W., Kasper, R., 2011. Development of a row guidance system for an autonomous robot for white asparagus harvesting. *Comput. Electron. Agric.* 79 (2), 216–225. <https://doi.org/10.1016/j.compag.2011.10.002>.
- Du, H., Qin, H., 2004. Medial axis extraction and shape manipulation of solid objects using parabolic PDEs. *Proceedings of the Ninth ACM Symposium on Solid Modeling and Applications*. Eurographics Association, pp. 25–35.
- Eaton, R., Katupitiya, J., Siew, K.W., Howarth, B., 2008. Autonomous farming: Modeling and control of agricultural machinery in a unified framework. In: *2008 15th International Conference on Mechatronics and Machine Vision in Practice*, pp. 499–504. <https://doi.org/10.1109/MMVP.2008.4749583>.
- Foley, J.A., Ramankutty, N., Brauman, K.A., Cassidy, E.S., Gerber, J.S., Johnston, M., et al., 2011. Solutions for a cultivated planet. *Nature* 478 (7369), 337.
- Gerland, P., Raftery, A.E., Ševčíková, H., Li, N., Gu, D., Spoorenberg, T., et al., 2014. World population stabilization unlikely this century. *Science* 346 (6206), 234–237. <https://doi.org/10.1126/science.1257469>.
- Graf Plessen, M., 2019. Coupling of crop assignment and vehicle routing for harvest planning in agriculture. *Artificial Intelligence Agriculture* 2, 99–109. <https://doi.org/10.1016/j.aiaa.2019.07.001>.
- Guijarro, M., Pajares, G., Riomoros, I., Herrera, P.J., Burgos-Artizzu, X.P., Ribeiro, A., 2011. Automatic segmentation of relevant textures in agricultural images. *Comput. Electron. Agric.* 75 (1), 75–83. <https://doi.org/10.1016/j.compag.2010.09.013>.
- Gupta, G., 2011. Algorithm for image processing using improved median filter and comparison of mean, median and improved median filter. *Int. J. Soft Comput. Eng. (IJSCE)* 1 (5), 304–311.
- Hague, T., Marchant, J.A., Tillett, N.D., 2000. Ground based sensing systems for autonomous agricultural vehicles. *Comput. Electron. Agric.* 25 (1–2), 11–28. [https://doi.org/10.1016/S0168-1699\(99\)00053-8](https://doi.org/10.1016/S0168-1699(99)00053-8).
- Hague, T., Tillett, N.D., 2001. A bandpass filter-based approach to crop row location and tracking. *Mechatronics* 11 (1), 1–12. [https://doi.org/10.1016/S0957-4158\(00\)00003-9](https://doi.org/10.1016/S0957-4158(00)00003-9).
- Hammer, B., Singh, S., Bergerman, M., 2010. Improving orchard efficiency with autonomous utility vehicles. 2010 Pittsburgh, Pennsylvania, June 20–June 23, 2010, 1. <https://doi.org/10.13031/2013.29902>.
- Hamuda, E., Glavin, M., Jones, E., 2016. A survey of image processing techniques for plant extraction and segmentation in the field. *Comput. Electron. Agric.* 125, 184–199. <https://doi.org/10.1016/j.compag.2016.04.024>.
- Hassall, J., 2010. Future trends in precision agriculture: a look into the future of agricultural equipment. *Nuffield Australia Project* 906.
- He, B., Liu, G., Ji, Y., Si, Y., Gao, R., 2010. Auto recognition of navigation path for harvest robot based on machine vision. *Int. Conference Computer Comput. Technol. Agric.* 138–148 https://doi.org/10.1007/978-3-642-18333-1_19.
- Hellström, T., 2002. Autonomous navigation for forest machines. Pre-Study Report, Department of Computing Science, Umea University, Sweden.
- Hvilshøj, M., Bøgh, S., Skov Nielsen, O., Madsen, O., 2012. Autonomous industrial mobile manipulation (AIMM): past, present and future. *Ind. Robot: Int. J.* 39 (2), 120–135. <https://doi.org/10.1108/01439911211201582>.
- Intel-corporation, 2018. Intel Realsense D435 Camera. Retrieved September 10, 2019, from <https://www.intelrealsense.com/stereo-depth/>.
- Islam, M., Chowdhury, M., Li, H., Hu, H., 2018. Vision-based Navigation of Autonomous Vehicle in Roadway Environments with Unexpected Hazards. *ArXiv Preprint. ArXiv: 1810.03967*.
- Jolliffe, I.T., Cadima, J., 2016. Principal component analysis: a review and recent developments. *Philos. Trans. Roy. Soc. A: Math., Phys. Eng. Sci.* 374 (2065), 20150202. <https://doi.org/10.1098/rsta.2015.0202>.
- Jones, M.H., Bell, J., Dredge, D., Seabright, M., Scarfe, A., Duke, M., MacDonald, B., 2019. Design and testing of a heavy-duty platform for autonomous navigation in kiwifruit orchards. *Biosyst. Eng.* 187, 129–146. <https://doi.org/10.1016/j.biosystemseng.2019.08.019>.
- Karkee, M., Adhikari, B., Amatya, S., Zhang, Q., 2014. Identification of pruning branches in tall spindle apple trees for automated pruning. *Comput. Electron. Agric.* 103, 127–135. <https://doi.org/10.1016/j.compag.2014.02.013>.
- Kise, M., Noguchi, N., Ishii, K., Terao, H., 2001. Development of the agricultural autonomous tractor with an RTK-GPS and a FOG. *IFAC Proc. Vol.* 34 (19), 99–104. [https://doi.org/10.1016/S1474-6670\(17\)33120-8](https://doi.org/10.1016/S1474-6670(17)33120-8).
- Kodinariya, T.M., Makwana, P.R., 2013. Review on determining number of Cluster in K-Means Clustering. *Int. J.* 1 (6), 90–95.
- Kumar, N. S., Sadasivam, V., Prema, K., 2004. Design and simulation of fuzzy controller for closed loop control of chopper fed embedded DC drives. 2004 International Conference on Power System Technology, 2004. *PowerCon 2004*, 1, 613–617. <https://doi.org/10.1109/ICPST.2004.1460067>.
- Kurka, P.R.G., Díaz Salazar, A.A., 2019. Applications of image processing in robotics and instrumentation. *Mech. Syst. Sig. Process.* 124, 142–169. <https://doi.org/10.1016/j.ymssp.2019.01.015>.
- Lam, L., Lee, S.-W., Suen, C.Y., 1992. Thinning methodologies—a comprehensive survey. *IEEE Trans. Pattern Anal. Mach. Intell.* 14 (9), 869–885. <https://doi.org/10.1109/34.161346>.
- Lee, W.S., Alchanatis, V., Yang, C., Hirafuji, M., Moshou, D., Li, C., 2010. Sensing technologies for precision specialty crop production. *Comput. Electron. Agric.* 74 (1), 2–33. <https://doi.org/10.1016/j.compag.2010.08.005>.
- Li, M., Imou, K., Wakabayashi, K., Yokoyama, S., 2009. Review of research on agricultural vehicle autonomous guidance. *Int. J. Agric. Biol. Eng.* 2 (3), 1–16. <https://doi.org/10.3965/j.issn.1934-6344.2009.03.001-016>.
- Li, Y., Hu, Z., Hu, Y., Chu, D., 2018. Integration of vision and topological self-localization for intelligent vehicles. *Mechatronics* 51, 46–58. <https://doi.org/10.1016/j.mechatronics.2018.02.012>.
- Madhulatha, T.S., 2012. An overview on clustering methods. *IOSR J. Eng.* 2 (4), 719–725. <https://doi.org/10.9790/3021-0204719725>.
- Makem, J.E., Fogg, H.J., Mukherjee, N., 2020. Automatic Feature Recognition Using the Medial Axis for Structured Meshing of Automotive Body Panels. *Comput. Aided Des.* 124, 102845. <https://doi.org/10.1016/j.cad.2020.102845>.
- Malavazi, F.B.P., Guyonneau, R., Fasquel, J.-B., Lagrange, S., Mercier, F., 2018. LiDAR-only based navigation algorithm for an autonomous agricultural robot. *Comput. Electron. Agric.* 154, 71–79. <https://doi.org/10.1016/j.compag.2018.08.034>.
- Mede, T., Chambon, G., Hagenmuller, P., Nicot, F., 2018. A medial axis based method for irregular grain shape representation in DEM simulations. *Granular Matter* 20 (1), 1–11. <https://doi.org/10.1007/s10035-017-0785-7>.

- Niewola, A., 2020. Mobile robot 6D localization using 3D Gaussian mixture maps in GPS-denied environments. *IEEE Intell. Syst.* 1 <https://doi.org/10.1109/MIS.2020.3046138>.
- Nof, S.Y., 2009. *Springer handbook of automation*. Springer Science & Business Media.
- Nørremark, M., Griepentrog, H.W., Nielsen, J., Søgaard, H.T., 2008. The development and assessment of the accuracy of an autonomous GPS-based system for intra-row mechanical weed control in row crops. *Biosyst. Eng.* 101 (4), 396–410. <https://doi.org/10.1016/j.biosystemseng.2008.09.007>.
- Opiyo, S., Zhou, J., Mwangi, E., Kai, W., Sunusi, I., 2020. A Review on Teleoperation of Mobile Ground Robots: Architecture and Situation Awareness. *Int. J. Control Autom. Syst.* 1–24. <https://doi.org/10.1007/s12555-019-0999-z>.
- Ossama, O., Mokhtar, H.M.O., El-Sharkawi, M.E., 2011. An extended k-means technique for clustering moving objects. *Egyptian Inform. J.* 12 (1), 45–51. <https://doi.org/10.1016/j.eij.2011.02.007>.
- Otsu, N., 1979. A threshold selection method from gray-level histograms. *IEEE Trans. Syst., Man, Cybernet.* 9 (1), 62–66. <https://doi.org/10.1109/tsmc.1979.4310076>.
- Ponnambalam, V.R., Bakken, M., Moore, R.J.D., Glenn Omholt Gjevestad, J., Johan From, P., 2020. Autonomous Crop Row Guidance Using Adaptive Multi-ROI in Strawberry Fields. *Sensors* 20 (18), 5249. <https://doi.org/10.3390/s20185249>.
- Qifeng, Y., Daokui, Q., Fang, X., Fengshan, Z., Guojian, H., Mingze, S., 2019. Mobile robot motion control and autonomous navigation in GPS-denied outdoor environments using 3D laser scanning. *Assembly Automation* 39 (3), 469–478. <https://doi.org/10.1108/AA-02-2018-029>.
- Quigley, M., Conley, K., Gerkey, B., Faust, J., Foote, T., Leibs, J., et al., 2009. ROS: an open-source Robot Operating System. *ICRA Workshop on Open Source Software*, 3 (3.2), 5. Kobe, Japan.
- Radcliffe, J., Cox, J., Bulanon, D.M., 2018. Machine vision for orchard navigation. *Comput. Ind.* 98, 165–171. <https://doi.org/10.1016/j.compind.2018.03.008>.
- Sahba, F., Nourani, Z., 2016. Smart tractors in pistachio orchards equipped with RFID. *World Automation Congress (WAC) 2016*, 1–6. <https://doi.org/10.1109/WAC.2016.7582995>.
- Sergiyenko, O., Tyrsa, V., 2020. 3D Optical Machine Vision Sensors with Intelligent Data Management for robotic swarm navigation improvement. *IEEE Sens. J.* <https://doi.org/10.1109/JSEN.2020.3007856>.
- Serrano, A., de Diego, I.M., Conde, C., Cabello, E., 2010. Recent advances in face biometrics with Gabor wavelets: A review. *Pattern Recogn. Lett.* 31 (5), 372–381. <https://doi.org/10.1016/j.patrec.2009.11.002>.
- Shahapure, K.R., Nicholas, C., 2020. Cluster Quality Analysis Using Silhouette Score. In: 2020 IEEE 7th International Conference on Data Science and Advanced Analytics (DSAA), pp. 747–748. <https://doi.org/10.1109/DSAA49011.2020.00096>.
- Shalal, N., Low, T., McCarthy, C., Hancock, N., 2013. A review of autonomous navigation systems in agricultural environments. Retrieved from <http://eprints.usq.edu.au/id/eprint/24779>.
- Shen, L., Bai, L., 2006. A review on Gabor wavelets for face recognition. *Pattern Anal. Appl.* 9 (2–3), 273–292. <https://doi.org/10.1007/s10044-006-0033-y>.
- Shen, W., Jiang, Y., Gao, W., Zeng, D., Wang, X., 2016. Shape recognition by bag of skeleton-associated contour parts. *Pattern Recogn. Lett.* 83, 321–329. <https://doi.org/10.1016/j.patrec.2016.02.002>.
- Smyth, P., 1996. Clustering Using Monte Carlo Cross-Validation. *KDD-96 Proc.* 1 (2), 26–33.
- Søgaard, H.T., Olsen, H.J., 2003. Determination of crop rows by image analysis without segmentation. *Comput. Electron. Agric.* 38 (2), 141–158. [https://doi.org/10.1016/S0168-1699\(02\)00140-0](https://doi.org/10.1016/S0168-1699(02)00140-0).
- Stentz, A., Dima, C., Wellington, C., Herman, H., Stager, D., 2002. A system for semi-autonomous tractor operations. *Autonomous Robots* 13 (1), 87–104. <https://doi.org/10.1023/A:1015634322857>.
- Stoll, A., Kutzbach, H.D., 2000. Guidance of a forage harvester with GPS. *Precis. Agric.* 2 (3), 281–291. <https://doi.org/10.1023/A:1011842907397>.
- Subramanian, V., Burks, T.F., 2005. Autonomous path navigation in citrus groves using machine vision and laser radar. 2005 ASAE Annual Meeting 1 <https://doi.org/10.13031/2013.19883>.
- Sugar, C.A., James, G.M., 2003. Finding the number of clusters in a dataset: An information-theoretic approach. *J. Am. Stat. Assoc.* 98 (463), 750–763. <https://doi.org/10.1198/01621450300000666>.
- Sun, B., Yearly, M., Sigmarsson, H.H., McDaniel, J.W., 2019. Fine Resolution Position Estimation Using Kalman Filtering. In: 2019 IEEE International Instrumentation and Measurement Technology Conference (I2MTC), pp. 1–5. <https://doi.org/10.1109/I2MTC.2019.8826857>.
- Takai, R., Barawid Jr, O., Ishii, K., Noguchi, N., 2010. Development of crawler-type robot tractor based on GPS and IMU. *IFAC Proc. Vol.* 43 (26), 151–156. <https://doi.org/10.3182/20101206-3-JP-3009.00026>.
- Takai, R., Yang, L., Noguchi, N., 2013. Development of Crawler-Type Robot Tractor Based on GNSS and IMU. *IFAC Proc. Vol.* 46 (4), 95–98. <https://doi.org/10.3182/20130327-3-JP-3017.00024>.
- Tan, L., J. J., 2013. Image Processing Basics. In *Digital Signal Processing* (Second Edn, pp. 683–765). Boston, Academic Press.
- Tian, K., Li, J., Zeng, J., Evans, A., Zhang, L., 2019. Segmentation of tomato leaf images based on adaptive clustering number of K-means algorithm. *Comput. Electron. Agric.* 165, 104962. <https://doi.org/10.1016/j.compag.2019.104962>.
- Tillett, N.D., Hague, T., 1999. Computer-vision-based hoe guidance for cereals—an initial trial. *J. Agric. Eng. Res.* 74 (3), 225–236. <https://doi.org/10.1006/jaer.1999.0458>.
- Tillett, N.D., Hague, T., Miles, S.J., 2002. Inter-row vision guidance for mechanical weed control in sugar beet. *Comput. Electron. Agric.* 33 (3), 163–177. [https://doi.org/10.1016/S0168-1699\(02\)00005-4](https://doi.org/10.1016/S0168-1699(02)00005-4).
- Tsogkas, S., Dickinson, S., 2017. Amat: Medial axis transform for natural images. In: *Proceedings of the IEEE International Conference on Computer Vision*, pp. 2708–2717.
- Umargono, E., Suseno, J. E., Gunawan, S. K. V., 2020. K-Means Clustering Optimization Using the Elbow Method and Early Centroid Determination Based on Mean and Median Formula. The 2nd International Seminar on Science and Technology (ISSTEC 2019), 121–129. <https://doi.org/10.2991/assehr.k.201010.019>.
- Underwood, J.P., Hung, C., Whelan, B., Sukkari, S., 2016. Mapping almond orchard canopy volume, flowers, fruit and yield using LiDAR and vision sensors. *Comput. Electron. Agric.* 130, 83–96. <https://doi.org/10.1016/j.compag.2016.09.014>.
- Van der Ploeg, J.D., 2000. Revitalizing agriculture: farming economically as starting ground for rural development. *Sociologia Ruralis* 40 (4), 497–511. <https://doi.org/10.1111/1467-9523.00163>.
- Vans, E., Vachkov, G., Sharma, A., 2014. Vision based autonomous path tracking of a mobile robot using fuzzy logic. *Asia-Pacific World Congress Comput. Sci. Eng.* 1–8. <https://doi.org/10.1109/APWCCSE.2014.7053862>.
- Wang, A., Zhang, W., Wei, X., 2019. A review on weed detection using ground-based machine vision and image processing techniques. *Comput. Electron. Agric.* 158, 226–240. <https://doi.org/10.1016/j.compag.2019.02.005>.
- Williams, H.A.M., Jones, M.H., Nejati, M., Seabright, M.J., Bell, J., Penhall, N.D., et al., 2019. Robotic kiwifruit harvesting using machine vision, convolutional neural networks, and robotic arms. *Biosyst. Eng.* 181, 140–156. <https://doi.org/10.1016/j.biosystemseng.2019.03.007>.
- Xue, J., Xu, L., 2010. Autonomous agricultural robot and its row guidance. 2010 International Conference on Measuring Technology and Mechatronics Automation, 1, 725–729. <https://doi.org/10.1109/ICMTMA.2010.251>.
- Xue, J., Zhang, L., Grift, T.E., 2012. Variable field-of-view machine vision based row guidance of an agricultural robot. *Comput. Electron. Agric.* 84, 85–91. <https://doi.org/10.1016/j.compag.2012.02.009>.
- Yamamoto, S., Yukumoto, O., Matsuo, Y., 2001. Robotization of agricultural vehicles—various operation with tilling robot. *IFAC Proc. Vol.* 34 (19), 203–208. [https://doi.org/10.1016/S1474-6670\(17\)33137-3](https://doi.org/10.1016/S1474-6670(17)33137-3).
- Yang, B., Yao, J., Guo, X., 2018. DMAT: Deformable medial axis transform for animated mesh approximation. *Comput. Graphics Forum* 37 (7), 301–311. <https://doi.org/10.1111/cgf.13569>.
- Yasseen, Z., Verroust-Blondet, A., Nasri, A., 2016. Shape matching by part alignment using extended chordal axis transform. *Pattern Recogn.* 57, 115–135. <https://doi.org/10.1016/j.patcog.2016.03.022>.
- Yoshizawa, S., Belyaev, A.G., Seidel, H.-P., 2003. Free-form skeleton-driven mesh deformations. In: *Proceedings of the Eighth ACM Symposium on Solid Modeling and Applications*, pp. 247–253. <https://doi.org/10.1145/781606.781643>.
- Zaitoun, N.M., Aqel, M.J., 2015. Survey on image segmentation techniques. *Procedia Comput. Sci.* 65, 797–806. <https://doi.org/10.1016/j.procs.2015.09.027>.
- Zhang, D., Lu, G., 2004. Review of shape representation and description techniques. *Pattern Recogn.* 37 (1), 1–19. <https://doi.org/10.1016/j.patcog.2003.07.008>.
- Zhang, D., Wong, A., Indrawan, M., Lu, G., 2000. Content-based image retrieval using Gabor texture features. *IEEE Trans. Pami* 13.
- Zhang, J., Tan, T., 2002. Brief review of invariant texture analysis methods. *Pattern Recogn.* 35 (3), 735–747. [https://doi.org/10.1016/S0031-3203\(01\)00074-7](https://doi.org/10.1016/S0031-3203(01)00074-7).
- Zhang, Q., 2013. Opportunity of robotics in specialty crop production. *IFAC Proc. Vol.* 46 (4), 38–39.